

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION
DURATION: 3 HOURS

SUMMER SEMESTER, 2022-2023
FULL MARKS: 150

CSE 4403: Algorithms

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 6 (six) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. You need to buy a cow for the Goru Party and have N markets to choose from, reachable from IUT. Each market is connected to IUT by a unique path with bi-directional roads. There is a fixed transportation fee collected at some stops on the route. Your goal is to calculate how many markets you can reach without exceeding the budget for the transportation fee M , provided the information about which stops on the way collect the transportation fee.

- | | |
|---|----------------------|
| a) Analyze the above-mentioned problem and propose a suitable algorithm to solve it. | 10
(CO3)
(PO3) |
| b) Justify the reason for choosing your solution. | 5
(CO2)
(PO2) |
| c) Formulate the worst-case time complexity for your proposed algorithm. | 3
(CO2)
(PO2) |
| d) Will your solution still work if the path between IUT and the market is not unique? If your answer is yes, justify. Otherwise, propose the modification required in your previous solution to solve the updated problem. | 7
(CO3)
(PO3) |

2. For the Goru Party, a number of dishes are planned to cook. Each dish has its own cooking duration and must be prepared within a fixed timeframe (i.e., each dish has a fixed start and end time). With just one cook who can handle one dish at a time, your aim is to determine the maximum number of dishes you can cook within their time constraints.

Now, answer the following questions:

- | | |
|---|-------------------------|
| a) Design a Greedy Algorithm to solve the problem described above and estimate its worst runtime complexity. | 8 + 2
(CO3)
(PO3) |
| b) Prove the optimality of your proposed algorithm. | 5
(CO1)
(PO1) |
| c) Design a Greedy Algorithm to calculate the minimum number of chefs required, if you want to cook all of the dishes. Estimate the worst runtime complexity. | 8 + 2
(CO3)
(PO3) |

3. Students want to decorate IUT using lighting during the Goru Party. For easier maintenance, the whole IUT campus is divided into multiple sectors. Given the cable required to connect different sectors, your goal is to calculate the shortest amount of cable required to connect all the sectors.
- Mention two algorithms, which can be used to solve the problem described above and estimate their worst runtime complexity. 6
(CO2)
(PO2)
 - Write a suitable algorithm to solve the problem described above. 7
(CO1)
(PO1)
 - Consider that, IUT is divided into seven sectors. The amount of cable required to connect various sectors is represented by 3 integer values, where the first 2 integers indicate the sector number and the third integer indicates the amount of required cable to connect them. Assume that the following values are given:

1 2 6
 1 4 4
 2 3 7
 2 4 8
 2 5 6
 3 5 4
 4 5 14
 4 6 5
 5 6 7
 5 7 8
 6 7 10

 Apply your selected algorithm to solve the problem for the aforementioned scenario. 12
(CO3)
(PO3)
4. Some sweets are planned to be served after the Goru Party. You have a line of sweet boxes, each containing a number written on top of it, indicating how many sweets you can take. Some boxes have negative numbers, meaning instead of taking, you must deposit sweets. Your task is to find the maximum number of sweets you can collect by choosing consecutive boxes strategically. For example: if the numbers on top of the boxes are: [-2, -1, 5, 0, -1, 3, 4, -2], then the maximum number of sweets you can collect is 11, which can be obtained by choosing the 3rd to 7th boxes. Now answer the following questions:
- Explain the naive brute force solution for the problem and discuss why this solution is not practical. 5
(CO3)
(PO3)
 - Design a solution using the divide and conquer approach to solve the problem. 10
(CO3)
(PO3)
 - Estimate the runtime of your solution using the recurrence relation and verify your result using the master theorem. 5
(CO1)
(PO1)
 - Is the divide-and-conquer solution the most efficient solution for this problem? If your answer is yes, justify. Otherwise, provide a better solution. 5
(CO3)
(PO3)

5. After the Goru Party, outgoing students are giving out candies standing in a line. You can not take candy from two consecutive people. Your task is to calculate the maximum total candy you can collect provided the number of candies each person will give. For instance, if there are 5 people with candies: [3, 10, 3, 1, 2], you can collect a maximum of 12 candies.

a) Explain why this problem can be solved using dynamic programming.

5
(CO1)
(PO1)

b) Design a top-down or bottom-up algorithm for solving this problem.

10
(CO3)
(PO3)

c) Determine the worst-case time complexity for your proposed algorithm.

3
(CO2)
(PO2)

d) Construct a memorization table for the following input: [1, 2, 3, 4, 5, 1, 2, 3, 4, 5].

7
(CO2)
(PO2)

6. a) What is Monte-Carlo simulation? Write down the pros and cons of this method. Using Monte-Carlo simulation estimate the value of π .

3 + 5
(CO1)
(PO1)

b) Show the steps of finding Huffman Encoding for the string representing your full name.

7
(CO3)
(PO3)

c) Apply the Ford-Fulkerson algorithm to the network flow graph provided in the Figure 1 to find the maximum flow possible from node S to T . Show each step of your operation.

10
(CO3)
(PO3)

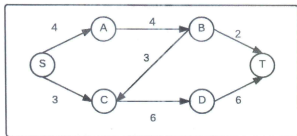


Figure 1: A network flow graph for Question 6.c

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MID SEMESTER EXAMINATION
DURATION: 1 HOUR 30 MINUTES

SUMMER SEMESTER, 2022-2023
FULL MARKS: 75

CSE 4403: Algorithms

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Answer all 3 (three) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) Calculate the worst time complexity using Big O notation of the following algorithms:

2 × 5
(CO1)
(PO1)

- i. Breath-First Search algorithm
- ii. Floyd-Warshall's algorithm
- iii. Bellman Ford's algorithm
- iv. Prim's algorithm
- v. Kruskal's algorithm

- b) Analyze the following code segments and find the upper bound runtime in Big O notation.

3 × 2
(CO1)
(PO1)

```
1 queue<int> q;
2 for (int i = 0; i < N; i++)
3     for (int j = 0; j < 10; j++)
4         q.push(i*j);
```

Code Snippet 1: A code snippet for Question 1.b)i

```
ii.
1 for (int i = 0; i < N; i++)
2     for (int j = 0; j < 10*N; j++)
3         cout << "Hello";
```

Code Snippet 2: A code snippet for Question 1.b)ii

```
iii.
1 int ar[N];
2 for (int i = 0; i < N; i++)
3     cin >> ar[i];
4 sort(ar, ar+N);
5 for (int i = 0; i < 10; i++)
6     for (int j = 0; j < N; j++)
7         cout << ar[j];
```

Code Snippet 3: A code snippet for Question 1.b)iii

- c) Calculate the space requirement for Counting sort applied to sort an integer (8 bytes) array of size 1000.

3 + 6
(CO1)
(PO1)

Consider the array given in Table 1, where XX = the last 2 digits of your student ID. Show the step-by-step sorting procedure of this array using radix sort.

Table 1: An array for Question 1.c

12	23	5	18	53	XX
----	----	---	----	----	----

2. a) There are n cities and m roads in a country. Each road connecting two cities is bi-directional. The cost to travel through each road may vary. Each city has a tourist fee. Every time a tourist visits a city, they need to pay the fee for that city. A group of three tourists start their journey from city 1 and want to finish their journey at city n . None of the tourists are allowed to use the same path to reach the destination. We consider two paths as different if they differ by at least one vertex. Given the description of each road (connecting cities and cost) and the tourist fee of each city, your goal is to find the most optimal travel paths for all the tourists to reach the destination minimizing the total expense of the group.
- Design a suitable algorithm to solve the problem. 10
(CO3)
(PO3)
 - Justify why your proposed solution will find the optimal solution. 5
(CO2)
(PO2)
 - Calculate the worst time complexity of your proposed solution. 3
(CO3)
(PO1)
- b) Will your solution still work, if the cities provide rewards for the tourists instead of tourist fees? If your answer is yes, justify. Otherwise, propose the solution for the updated scenario. 7
(CO2)
(PO2)
3. a) Explain how path compression is used to flatten the tree in union-find operation. Apply Kruskal's algorithm to the graph provided in Figure 1 to find the minimum spanning tree. Show the content of the parent array in each step of the union-find operation. 10
(CO2)
(PO2)



Figure 1: A directed graph for Question 3.a

- b) The intra-department cricket tournament of IUT is going to be held. A team gets 2 points if they win a match and they get 1 point if they lose a match. In case of a tie, a coin toss is used to decide the winner. Given the target point (P), your goal is to calculate in how many different ways a team can obtain the target points.
- For example: If the target point $P = 4$, then the team can obtain the point in 5 different ways:
- L L L L
 - L L W
 - L W L
 - W L L
 - W W
- Explain whether this problem can be solved using Dynamic Programming (DP) or not. Justify your answer. 5
(CO3)
(PO1)
 - Design a bottom-up algorithm for solving this problem and find the initial values for the DP table. 5
(CO3)
(PO3)
 - Formulate the worst-case time complexity for your proposed algorithm and construct a DP table for $P = 10$. 5
(CO3)
(PO2)



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MID SEMESTER EXAMINATION

SUMMER SEMESTER, 2021-2022

DURATION: 1 HOUR 30 MINUTES

FULL MARKS: 75

CSE 4403: Algorithms

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Answer **all 3 (three)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

1. The cost of flagship mobile phones is significantly higher in Bangladesh than in Singapore. Mr. Gerald, who is traveling from Singapore to Bangladesh next month, has devised a profitable business plan. As he is a greedy individual, he intends to purchase flagship phones in Singapore and sell them in Bangladesh for a profit. 25
(CO3)
(PO1)

For this purpose, Mr. Gerald goes to a local tech store in Singapore. There are N phones on display. For each phone i ($1 \leq i \leq N$), he has the following information available to him:

- price in USD (p_i),
- price in BDT (q_i)
- type (t_i).

Assuming $1 \text{ USD} = 105 \text{ BDT}$, it is guaranteed that $q_i > 105 * p_i$ for each phone i . There are two types of phones: made by Apple (type 0) and made by Samsung (type 1).

Gerald has a budget of P USD. He can buy as many phones of any type as he likes as long as it does not exceed his budget. But he must ensure to buy at least K ($K \leq N$) Samsung phones as their demands are higher. Gerald wants to know the maximum profit he can achieve while ensuring all the constraints.

Propose a Dynamic Programming solution for this maximization problem under some constraints. You need to define a set of subproblems, relate the subproblems recursively, provide base cases, construct a solution from the subproblems, and analyze the running time.

2. a) Apply DFS on the following graph and show the order in which order the new vertices are visited. Assume that both the vertices in the adjacency list and neighbors of each individual vertex are ordered alphabetically. 12
(CO1)
(PO1)

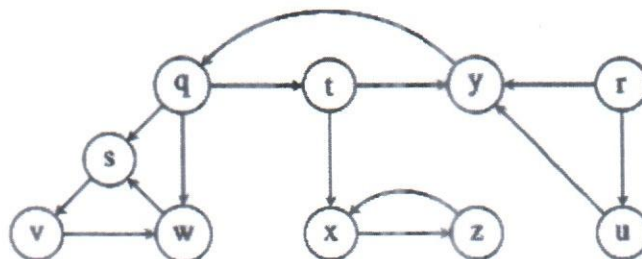


Figure 1: Graph for Question 2(a)

- b) Asses the pros and cons of using adjacency matrix and adjacency list. 8
(CO2)
(PO2)
- c) Give a real-life example of a graph where traversing all edges are required. 5
(CO3)
(PO1)

3. a) The Omicron variant of the Covid-19 virus was initially detected in South Africa, where it had infected many people. Some people infected with the virus recently arrived in Bangladesh from South Africa. When they arrived in Bangladesh, some of them moved to the Bashundhara Residential Area and started living in various buildings. The virus is infectious, and it infects everyone in a building when one infected person enters. Furthermore, as each day passes, the neighboring building is compromised as well (i.e., if at day 1, building **A** is infected, then at day 2, the virus will be spread at all the neighboring (east, west, north, and south) buildings of **A**). Given a grid map of Bashundhara Residential Area, where each cell with the letter '**X**' indicates an Omicron-infected building and a cell with the letter '**Y**' represents an uninfected building.

13
(CO3)
(PO3)

Devise an efficient algorithm to determine the number of days it will take the Omicron virus to infect the whole residential area. An example is given for better understanding:

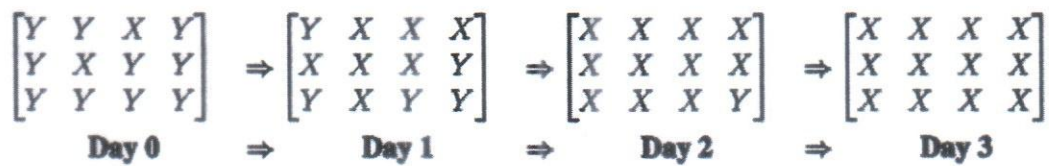


Figure 2: Example demonstration for Question 3(a)

The initial grid contained just two buildings with Omicron infected persons on Day 0, but by the end of Day 3, all the buildings in the grid had been contaminated. Thus, the deadly virus had spread across the residential area in just **three** days.

- b) **Speed in Need** is a racing video game set in the fictional country, Little America. There are N towns in Little America. The towns are connected by M bidirectional roads. Each town has a unique positive integer difficulty level, $diff$. When you go from town u to town v , you will face obstacles if $diff[u] < diff[v]$. Your car is currently in town X . Assume that the difficulty level of town X is the highest among all.

12
(CO2)
(PO2)

Determine whether you can traverse all the towns starting from X without facing any obstacle. If so, find the traversal order of the towns with proper justification.

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SEMESTER FINAL EXAMINATION

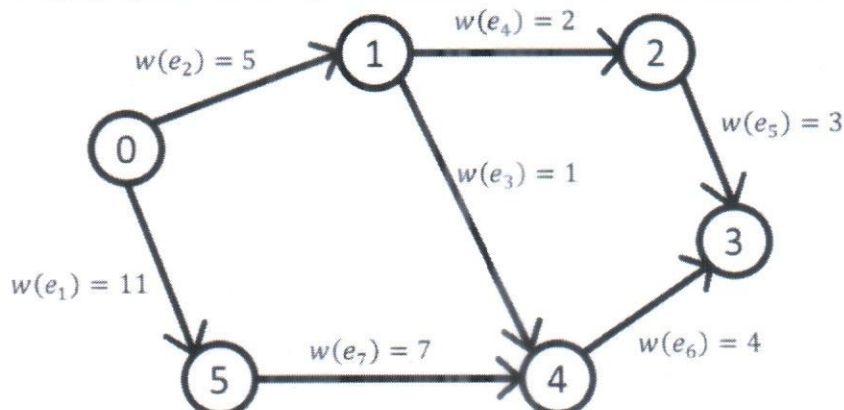
SUMMER SEMESTER, 2020-2021

DURATION: 3 HOURS

FULL MARKS: 150

CSE 4403: Algorithms**Programmable calculators are not allowed. Do not write anything on the question paper.**Answer all **6 (six)** questions. Marks of each question and corresponding CO and PO are written in the right margin with brackets.

1. a) Answer the following questions: 3+5+5
- Find the worst-case complexity of counting sort by explaining its working principle. (CO1)
(PO1)
 - Calculate the memory requirement of counting sort when sorting 32 bits and 64 bits numbers.
 - Explain how radix sort with counters overcomes the memory requirement issues of counting sort by using the idea of prefix sum.
- b) Show the steps of sorting the following list of numbers using radix sort with counters: 12
- 12 23 25 14 36 (CO2)
(PO1)
2. a) Analyse the configuration of each of the following graphs and choose the most suitable algorithm among Dijkstra, Bellman-Ford and Floyd-Warshall to find the shortest path. 5x3
- You are provided with an undirected graph with negative weights. Your goal is to find the shortest path between a pair of vertices. (CO1)
(PO2)
 - A directed weighted graph with negative weights, but with no cycle is provided as input and your goal is to find the shortest path for all possible pairs of vertices.
 - A directed disconnected graph with negative cycles is provided as input. Your goal is to detect whether there exists a negative cycle or not.
- Justify the reason for your selection. Also, provide the worst-case complexity for your selected algorithm.
- b) Consider the graph presented in Figure 1 and answer the following questions:

**Figure 1:** An undirected weighted graph

- Find the order of the vertices to be deleted from the priority queue if you apply Dijkstra's algorithm with vertex 0 as the source. 4
(CO1)
(PO1)
- Apply the Bellman-Ford algorithm to find the shortest path from vertex 0 to vertex 4. 6
(CO2)
(PO1)

3. a) Assume you have obtained a Minimum Spanning Tree (MST) T from graph G . Determine whether the tree T will remain MST if you decrease the weight of one of the edges $e \notin T$. Justify if your answer is yes. If your answer is no, then provide an efficient algorithm to find the updated MST. 10 (CO2) (PO1)
- b) Consider the graph presented in Figure 1 as an undirected graph and apply Prim's algorithm starting from vertex 1 on the graph to find the minimum spanning tree. 10 (CO2) (PO1)
- c) An MST edge whose deletion from the graph would cause the MST weight to increase is called a critical edge. Find all critical edges of the undirected version of the graph presented in Figure 1. 5 (CO1) (PO1)
4. a) Write down an algorithm to approximate the value π (π) using Monte Carlo simulation. 6 (CO1) (PO1)
- b) Assume a box contains an infinite supply of red and green balls. You are playing a game, where you randomly pick a ball from the box and if two consecutive red balls are picked, then the game is over. The score of the game is determined by the number of balls picked before the game is over. The probability of picking the green ball is p .
- i. Derive a mathematical formula to calculate the expected score of the game. 5 (CO2) (PO1)
- ii. Propose an algorithm to find the expected score of the game using the Monte Carlo simulation. 7 (CO2) (PO3)
- c) Explain by providing a Las Vegas algorithm for the solution of the 8-Queen problem. How does the Las Vegas algorithm finds the exact result unlike approximating the result as Monte Carlo simulation? 7 (CO1) (PO2)
5. a) The following is a general recurrence relation for a divide and conquer algorithm:

$$T(n) = a T(n/b) + f(n)$$
3+3+6 (CO1) (PO1)
- i. Explain whether is it necessary to have the value of b and a greater than 1 or not.
- ii. Determine the value of a , b and $f(n)$ for the merge sort
- iii. Using the recurrence tree, determine the runtime for the following recurrence relation:

$$T(n) = T(n/3) + T(2n/3) + cn$$
- b) Given a set of points in the plane, the **convex hull** of the set is the smallest convex polygon that contains all the points of it.
- i. Propose a divide and conquer algorithm to find the convex hull from a given set of points. 10 (CO2) (PO2)
- ii. Determine the runtime of your proposed algorithm. 3 (CO3) (PO3)
6. a) To manage the booking of the IUT auditorium, a Google Form is circulated among the staff. The staff choose an opening and closing time based on their need. Given all the booking requests for a particular day, your job is to create a schedule so that the maximum number of staff can be served.
- i. Propose a greedy algorithm to solve this problem. 10 (CO1) (PO1)
- ii. Find the sorting criteria of the request while processing, to obtain the global optimal result. Justify your answer. 8 (CO2) (PO3)

- b) Consider the flow network graph provided in Figure-2:

7
(CO1)
(PO1)

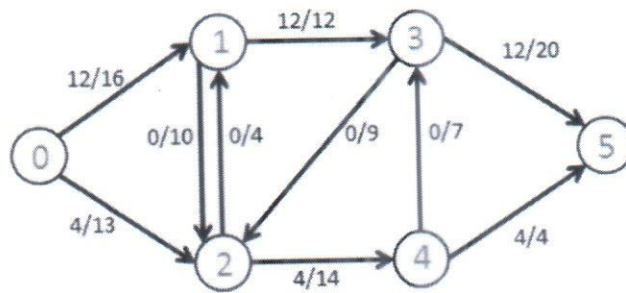


Figure 2: A flow network graph

Find the maximum flow over the network by applying the Ford-Fulkerson algorithm, considering vertex 0 as the source and vertex 5 as the sink.

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Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION

SUMMER SEMESTER, 2018-2019

DURATION: 3 Hours

FULL MARKS: 150

CSE 4403: Algorithms

Programmable calculators are not allowed. Do not write anything on the question paper.

There are **8 (eight)** questions. **Question no 2,3 & 4 are mandatory to answer.**

Answer any **3 (three)** from the remaining.

Figures in the right margin indicate marks.

1. a) Friday has many distinguishing features and virtues that Allah has bestowed upon this day and not others and so it is considered as the most important day of the whole week. There are a number of different Sunnah of Prophet Muhammad(saw) on this day. For example, on this day Muslims are encouraged to go to the mosque early on foot and sit near Imam. It's also said that whoever reads Surah Al-Kahf on Friday makes him illuminated with light between the two Fridays. Before going to the mosque, everyone should finish some activities like cutting nails, taking a bath and then go to the mosque wearing clean clothes. Applying perfume, doing miswaak are also important Sunnah before coming to the mosque. In addition, every Muslim should make a lot of Dua and read durood on Prophet Muhammad(saw) on this day. Based on the above-mentioned facts, find out the correct order of performing these Sunnahs and show the dependency of the activities (Sunnahs) using DAG in a linear fashion where all the directed edges go from left to right. 18
- b) Suppose a set of Strongly Connected Components (SCC) were found from a directed graph. Afterward, an edge was added into the graph and the procedure was performed again, but it resulted in no change in the set of SCC. What are the possible positions for the inclusion of that edge? 7

[Mandatory]

2. a) Showing proper argument, establish the fact whether Dijkstra's algorithm for solving the single-source shortest-paths problem on a weighted directed graph $G = (V, E)$ follows Dynamic Programming or Greedy approach. 10
- b) Free medicines are being distributed in a camp where patients are sitting maintaining a line. The medical officer has examined their health status and given a rating based on the necessity of drugs. A patient with a higher rating means s/he is in the bad physical condition and requires more medicine. 15
 A nurse is distributing the medicines based on rating and she is instructed to give at least one strip of a tablet to each patient. If two patients are sitting next to each other, the one with the higher rating must get at least one more strip of medicines. Again, if two patients with equal rating sitting next to each other will get the same number of medicines. The nurse is given a goal to minimize the overall number of strips of medicine given to the patients.
 For example, if the ratings of the patients are (2,5,5) the distribution of medicines should be (1,2,2) that is 5 strips in total. Again, if the ratings are (4,6,6,10,10,1,4,2,1) the distribution should be (1,2,2,3,3,1,2,2,1) that is 17 strips in total.
 Analyze the scenario and characterize the structure of the optimal solution. Show whether this problem can be further divided into subproblems or not. Finally, propose an algorithm to find out the optimal solution in time-complexity $\theta(n)$, where n is the total number of patients.

[Mandatory]

3. a) Supposing that a data file contains a sequence of 3-bit characters such that all possible characters in the system are about equally common: the maximum character frequency is less than twice the minimum character frequency. 10

Based on the facts mentioned above, justify the following claim, "Huffman coding, in this case, is efficient than using an ordinary 3-bit fixed-length code."

- b) Suppose an edge (u, v) exist in a directed graph. After applying Depth-first-search (DFS) on the graph; the discovery time and finishing time for every vertex are stored using the attributes d & f respectively. 15

Match the two sides of the following table by showing appropriate explanations.

Edge (u, v) is:	
A back edge if and only if	$v.d \leq u.d < u.f \leq v.f$
A forward edge or tree edge if and only if	$v.d < v.f < u.d < u.f$
A cross edge if and only if	$u.d < v.d < v.f < u.f$

[Mandatory]

4. a) Long-time ago, there was a village named Lake-valley. The place consisted of n lakes represented using the set $S = \{L_1, L_2, \dots, L_n\}$. Two types of fish lived in these lakes, but only one type in each lake. Both types of fish looked exactly the same, smelled exactly the same when cooked and had the exact same delicious taste – except one was poisonous and would turn any villager into flesh-craving zombies. This happened because the villagers angered a magician who cursed k lakes from those n lakes to have poisonous fish ($k \leq n$). Unfortunately, the only known test to determine the condition of the lake is to serve some of the fish to a human and see whether they turn into zombies or not! Several brave volunteers offered to undergo such an experiment, but they are only willing to try their luck once. Each volunteer is willing to eat a single serving of fish that mixes together fishes from $S' (\subseteq S)$ lakes, which reveals at least one lake in S' has poisonous fish. Your goal is to use the fewest possible volunteers in order to determine, for each lake L_i , whether the fish in it is poisonous or not. 15

Save the world by designing an algorithm to determine which k of the n lakes have poisonous fish using $O(k \log_2(n))$ experiments.

- b) Show that Merge sort requires $\theta(n \log(n))$ auxiliary space to sort n numbers which can further be optimized to $\theta(n)$. 10

5. a) Design an algorithm that will find the longest-path from a source vertex to every other vertex in a directed weighted graph that contains no cycles. The parameters of the function will be the graph(G), source(s) and set of vertices(v). For every vertex, the algorithm should show the longest distance from the source vertex along with the vertices comprising that path. Additionally, show the unreachable nodes from the source vertex. 10

- b) Explain the difficulty faced in single-source shortest-path problem if Negative Weights are allowed for the edges. How does the Bellman-Ford algorithm deal with this issue? 8

- c) Suppose the vertices of a directed weighted graph are represented using a min-priority queue(Q) and the Bellman-Ford algorithm is applied to find the shortest path from the source vertex to every other vertex. Explain the impact this will bring on the overall complexity of the algorithm. 7

6. a) Find an optimal parenthesizing of a matrix-chain product whose sequence of dimensions is $(4, 10, 3, 12, 20, 7)$ 15

- b) Let $X = (x_1, x_2, \dots, x_m)$ and $Y = (y_1, y_2, \dots, y_n)$ be sequences, and let $Z = (z_1, z_2, \dots, z_k)$ be any Longest common subsequence(LCS) of X and Y . Characterize the LCS and propose a recursive definition utilizing the optimal substructure of the problem. 10

7. a) Use the master method to give tight asymptotic bounds for the following recursions: 9
- i. $T(n) = 3T\left(\frac{n}{4}\right) + n\log(n)$
 - ii. $T(n) = T\left(\frac{n}{2}\right) + 2^n$
 - iii. $T(n) = 0.5T\left(\frac{n}{2}\right) + \frac{1}{n}$
- b) Assume that, instead of always selecting the first activity to finish, we select the last activity to start that is compatible with all previously selected activities. Describe how this approach is a greedy algorithm, and prove that it yields an optimal solution. 8
- c) Given an adjacency-list representation of a directed graph, how to compute the out-degrees and in-degrees of every vertex? What would be the case if the graph was represented using adjacency-matrix? What will be the time taken in both cases? 8
8. a) Build a Max-Heap from the following set of numbers $\{3, 6, 27, 17, 1, 5, 12, 7, 4, 0, 10, 8, 13, 9\}$ 10
- b) Show that, if a node in a Binary Search Tree has two children, then its successor has no left child and its predecessor has no right child. 8
- c) Prove the correctness of the Insertion-sort algorithm. 7

180

$O(\text{---})$ time. (Fill in the blank with justification.)

- vi. Determine the average processing time $T(n)$ of the following algorithm:

```
int myTest (int n) {  
    if (n <= 0) return 0;  
    else {  
        return myTest (n - 1);  
    }  
}
```

- b) Design a function which will take a node of a *Binary Search Tree* as input and show all of its *Ancestors*. The algorithm should be able to produce output even if the node doesn't have any ancestor or the node is not present in the tree. 7

[Mandatory]

4. Lately, Bill Gates is thinking of opening a restaurant! Being one of the smartest guys on the planet, obviously he is planning to digitalize it. Each customer will have a membership card which will have information about him/her along with the transactions made so far. Based on that information, a rating is given to the customers. 25

Whenever a customer enters the restaurant, s/he needs to scan the membership card by which the authority tracks who are present at the moment. A big screen is placed in front of the counter where the information is being shown about the current customer being served, the next customer to be served and the total number of customers present at the moment. While choosing the next customer to be served, the system looks for the highest rated customer present at that moment.

As this is Bill Gate's restaurant, a huge number of customers are expected to come every day! To handle the data, he wants to store it in the best possible way! His employees came up with different solutions to store the information so that the information on the big screen can be shown efficiently. Some of the proposed ways are:

- i. Array
- ii. Linked list
- iii. Heap
- iv. Binary Search Tree
- v. AVL Tree
- vi. Red Black Tree

Provide Arguments about the worst-case time complexity for each of the solutions to perform the required operations and choose the best one. Justify your choice.

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
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Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION

SUMMER SEMESTER, 2017-2018

DURATION: 3 Hours

FULL MARKS: 150

CSE 4403: Algorithms

Programmable calculators are not allowed. Do not write anything on the question paper.

There are **8 (eight)** questions. Answer any **6 (six)** of them.

Figures in the right margin indicate marks.

1. a) What is an algorithm? How does a software use different algorithms together to achieve a particular high level goal? 2+3
 b) Prove that the expected running time of quicksort using randomized-partition is $O(n \log n)$. 10
 c) The *maximum sub-array* problem and *Longest Increasing Subsequence (LIS)* problem have some similarities and yet they are different. Provide an analysis of the two problems and outline how they are similar and how they are dissimilar. You may draw diagrams if needed. 10
2. a) Although merge sort runs in $\theta(n \log n)$ worst-case time and insertion sort runs in $\theta(n^2)$ worst-case time, the constant factors in insertion sort can make it faster in practice for small problem sizes on many machines. Thus, it makes sense to coarsen the leaves of the recursion by using insertion sort within merge sort when subproblems become sufficiently small. Consider a modification to merge sort in which n/k sublists of length k are sorted using insertion sort and then merged using the standard merging mechanism.
 i. Show that insertion sort can sort the n/k sublists, each of length k in $\theta(nk)$ worst-case time.
 ii. Show how to merge the sublists in $\theta(n \log(n/k))$ worst-case time. 8
 b) Define the optimal substructure of Longest Common Subsequence (LCS) problem. 7
 c) Compute the LCS for the following two strings: 10
 ABABCD A and CDABAB.
3. a) What do you understand by 'optimal substructure'? Why is it a necessary property for a problem to hold if it is to be solved by dynamic programming? 7
 b) How can you find a cycle in a graph by the two set operations **Union**(u, v) and **Find-Set** (x)? 8
 c) Compute the *prefix function* (π) for a KMP matcher for the following pattern: 10
 P=abacbcabac
4. a) Prove that Breadth-First Search (BFS) provides the shortest path from the source to all other vertices. 7
 b) The algorithms for finding the Minimum Spanning Tree (MST) in a graph are greedy algorithms. What is the theoretical foundation for MST algorithms to be greedy? 8
 c) Prove that the height of a B-tree is $h \leq \log_2 \left(\frac{n+1}{2} \right)$ 10

5. a) Many of the practical algorithms are derived from common-sense. Mention two algorithms that are common sense (do not quote sorting algorithms) and how the algorithms are mapped to common sense. 8
- b) Analyze the complexity of Kruskal's MST algorithm. 7
- c) Insert the following keys into a B-tree (Assume $t=2$): 10

G M P X A C D E J K N O R S T U V Y Z

6. a) Activity selection problem can be solved by using a greedy algorithm. What is the greedy choice it makes in every iteration? How can you prove that the greedy activity selection algorithm will provide the optimal set of selected activities? 10
- b) A dynamic programming algorithm that matches two time series X and Y has the following cost function: 15

$$c[i, j] = \begin{cases} 0 & \text{if } i = 0 / j = 0 \\ d[i, j] + \min(c[i-1, j], c[i, j-1], c[i-1, j-1]) & \text{otherwise} \end{cases}$$

Where $d[i, j] = |X[i] - Y[j]|$

Calculate the minimum matching cost and find the minimal matching path (warp path) between the two time series as given below:

$X = [1, 2, 3, 7, 2, 4, 5]$ $Y = [1, 1, 6, 2, 3, 3, 4, 5]$

7. a) Show the solution for the following recurrence: 9
- $$T(n) = 2T(\sqrt{n}) + \log n$$
- b) Analyze the complexity of Build-Max-Heap() algorithm. 10
- c) Briefly describe the O , θ and Ω notations. 6
8. a) How do you analyze the correctness of an algorithm? 5
- b) Use Floyd Warshall algorithm to calculate the all pair shortest path using the following adjacency matrix for 4 vertices: 10

$$\begin{pmatrix} 0 & 3 & 8 & \infty \\ \infty & 0 & 3 & 1 \\ \infty & 4 & 0 & 2 \\ 2 & \infty & -5 & 0 \end{pmatrix}$$

- c) Use graham scan algorithm to find the convex hull of the following points: 10
- $(1, 3), (1, 2), (-1, -1), (4, 4), (0, 1), (8, 8), (2, 7), (4, 3), (3, 4).$